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Inventor(s) Adelmand; Hemin


## Mechanical Fitting


### Abstract

The present invention proposes a mechanical fitting for connecting pipes or for attaching a pipe, a method of assembling such a fitting and a pipe, and a tool for the assembly of a fitting and a pipe. The mechanical fitting comprises a main body (**10, 210**) extending axially from an open end (**11**) and defining an insertion space for a pipe, a clip ring (**30, 130, 230**) designed to close over a pipe, and a nut (**40, 240**) movably mounted at least in part into the main body for tightening the clip ring. The clip ring (**30; 130**) is arranged inside the main body for being contactable by the nut (**40; 240**) and has an inner circumferential boss or protrusion (**32; 132**), to be clipped into a corresponding outer circumferential groove (**4**) of the pipe (**2**).



**Inventors:** **Adelmand; Hemin (Élancourt, FR)**

**Applicant:** **Aliaxis Research & Technology** (Élancourt, FR)



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present invention claims priority to the French Patent Application 24/02205 filed on 5 Mar. 2024. The French Patent Application 24/02205 is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates to a mechanical fitting for connecting a pipe, and method of assembling such a mechanical fitting and a pipe, and a tool usable in such a method.

BRIEF DESCRIPTION OF THE RELATED ART

[0003] Press fittings are used to connect not only metal but also plastic or composite metal-plastic pipes. They are used to connect appropriately arranged sections of pipe, with their connecting pieces being inserted into the ends of the pipes to be connected, where they are then deformed or crimped or pressed. They are fastened in certain zones, e.g. using press sleeves on the pieces by using so-called system compression tools that usually have interchangeable pressing jaws.

[0004] The pipe, tubes, or conduit systems are assembled in several steps: first the pipes are inserted into a fitting, and then the fitting is crimped to guarantee good mechanical strength and a leak-tight installation. The compression is essentially a pressure on the fitting sleeve-also referred to as a press sleeve-which causes the deformation or shaping of the fitting to the pipe which is inserted into it. The pressure is usually applied to the fitting using a press jaw acting as a compression tool. These press jaws apply radial pressure to the fitting material in order to bring an interaction with the surface of the pipe(s)/pipe(s) to be connected, thus achieving a reliable connection and a reliable seal.

[0005] Other fittings, such as used for example for hoses such as in garden appliances, use a threaded engagement to provide for the crimping of the pipe and rely on the fact that a part of the fitting engages with the pipe.

[0006] In both cases, the connection relies on a deformation of the pipe to be connected, in most cases using an insert inside of the tube to be connected.

[0007] For different pipe arrangements, press fittings can be designed as curved, angular or T-shaped pieces, with commonly one, two or three connection pieces being provided.

[0008] The pipe or pipe can be mono material and made of thermoplastic materials including but not limited to PE, PB, PEX, PERT, PER . . . or Composite (multi material pipe structure) with different layers of functional materials (PEX, PERT, PVDF, Aluminium . . . ).

[0009] It is known to use full plastic press fittings or press fittings having a fitting body made of metal (brass, Stainless steel . . . ), or thermoplastic (PVC, PPSU, PVDF, PPS), that can also be reinforced with fibre glass reinforcement.

[0010] With the introduction of pipes in c-PVC material, however, compression fittings relying on deformation of the material of the pipe for the connection do not show good results. Indeed, c-PVC exhibits different properties and is more brittle than other materials.

[0011] One solution that was proposed is a solvent cement connection. However, the regulation is evolving, and it will be necessary to avoid the addition of solvents.

[0012] Therefore, it is necessary to find another solution for compression fitting, to avoid the use of solvent.

[0013] One object of the present disclosure is to propose a mechanical fitting that allows the connection of PVC based materials, such as c-PVC, PVC, and PVC-U piping systems for hot and cold-water installations, in particular pressurized installations.

SUMMARY OF THE INVENTION

[0014] These and other objects of the present invention are achieved by a mechanical fitting for connecting a pipe, comprising: a main body extending axially from an open end and defining an insertion space for a pipe, a clip ring designed to close over a pipe, and a nut movably mounted at least in part into the main body for tightening the clip ring. The clip ring is arranged inside the main body for being contactable by the nut and has an inner circumferential boss or protrusion, to be clipped into a corresponding outer circumferential groove of the pipe.

[0015] In an aspect, the boss has an insertion surface extending from a pipe insertion end to an inner radial boss surface followed by a pull-out surface, wherein the insertion surface forms an insertion angle with respect to the inner radial boss surface wider than a pull-out angle formed by the pull-out surface with respect to the to the inner boss surface. The geometry of the boss allows to redistribute the pullout force into axial and horizontal force, as well as facilitating insertion of the pipe in the clip ring.

[0016] The insertion surface and/or the pull-out surface may be provided with inner grooves. In an aspect, the clip ring has at least one support surface extending axially from the pull-out surface and/or from the insertion surface, forming at least one shoulders designed to lie against the pipe outer surface of the pipe when the pipe is inserted. This design allows diminishing the constraints or load on the pipe at the groove and transmit the constraints to the clip ring through the boss inserted in the groove.

[0017] The clip ring can be an open ring, wherein an inner diameter of the clip ring when open is larger than an outer diameter of the groove, preferably substantially corresponding to an outer diameter of pipe, and with an inner diameter of the clip ring when closed is equivalent to the outer diameter of the groove. This ensures that the clip ring closes and transmit the force on the pipe.

[0018] In an aspect, the clip ring has an outer conical shape, with an outer conical surface, with the diameter decreasing from the pipe insertion end to the fitting side, and wherein the clip ring is housed in a clip receiving part of the main body, the clip receiving part having a corresponding conical shape. A conical shape facilitates both the pipe insertion, and the force transmission during closing of the clip ring when the nut is moved into the fitting.

[0019] In an aspect, the nut has a first nut portion to be mounted inside a nut receiving part of the main body and a nut head, with the first nut portion having nut threads to be threaded into internal threads of the nut receiving part, and a nut head part extending from the first nut portion, the head part being provided to abut against the open end of the main body.

[0020] A joint can be provided inside a joint housing portion in the main body located adjacent the clip receiving part.

[0021] A at least one of these features is fulfilled: the nut is made of fibre reinforced polymer, polymer matrix, or composite, the main body is made of fibre reinforced polymer, polymer matrix, or composite, the ring is made of polymer, fibre reinforced polymer, polymer matrix, or composite.

[0022] The present invention further provides for a method of assembling of such a mechanical fitting and a pipe provided with an outer circumferential groove. The method comprises inserting the pipe into the fitting main body and the clip ring or inserting both the pipe with the clip ring mounted thereon into the fitting main body, preferably up to the inner radial rib, and moving the nut in the main body, by sliding and/or threading the nut in the main body to close the clip ring around the pipe. The method comprises inserting the boss of the clip ring into the groove of the pipe.

[0023] In an aspect, the groove is provided by a step of cutting the pipe to form the groove, preferably an outer circular groove on the pipe, preferably comprising the steps of forming simultaneously the outer circular circumferential groove on the pipe and chamfering the insertion end of the pipe and/or correcting the insertion end. A circular groove can be made prior to assembly, even if the pipe has some ovalisation. With other words, the groove is an ovalisation compensation tool. This is particularly advantageous when using PVC pipes which are brittle, and which cannot withstand deformation.

[0024] It should be noted that the closing of the nut is visual indication to good and tight

connection. In addition, after tightening, the clip ring is fully closed and there is no space for accumulation of impurities, between the joint, the rubber, the pipe preventing the growth of micro-organism, bio-film and accumulation of impurities.
[0025] It should be noted the clip ring is designed to adapt with the ovalisation pipe. The dimension of clip-ring is equivalent to the dimension of the groove, which allows compensating ovalisation or pipe removal materials. Additionally, the clip ring has an angled outer surface that allows the reduction of the clip ring diameter upon insertion into the fitting or upon tightening of the nut. The clip ring fills the groove space in the pipe and support the pipe under radial load.
[0026] The present invention further propose a tool usable in the above method, wherein the tool has a main hollow body to be mounted over the pipe, with a pipe stop to stop the pipe end, the tool having a groove cutter, at an axial distance from the pipe stop, wherein the axial distance between the pipe stop and the groove cutter can be adjusted to correspond to an axial distance between the pipe stop and the boss of the clip ring of the mechanical fitting.
[0027] In an aspect, the tool has at least one of a chamfering blade extending from the pipe stop for chamfering the pipe insertion end, and a corrective blade at the pipe stop for correcting the pipe end, a pair of arms, preferably opposite arms, extending outwardly from the main body.

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## Description

DESCRIPTION OF THE DRAWINGS
[0028] Other characteristics and advantages of the invention will be more clearly evident upon reading the description of several currently preferred embodiments, provided as examples only, with reference to the attached drawings, wherein:
[0029] FIG. **1** shows a mechanical fitting according to a first aspect of the present disclosure,
[0030] FIG. **2** shows the mechanical fitting of FIG. **1**, in an exploded view,
[0031] FIG. **3** shows the mechanical fitting of FIG. **1**,
[0032] FIG. **4** shows the mechanical fitting of FIG. **1** assembled with a pipe,
[0033] FIG. **5** shows a main body of the mechanical fitting according to present disclosure,
[0034] FIG. **6** shows a clip ring usable in a mechanical fitting according to an aspect of the present disclosure,
[0035] FIG. **7** shows a clip ring usable in a mechanical fitting according to an aspect of the present disclosure with a pipe,
[0036] FIG. **8** shows details of the clip ring of FIG. **7** according to an aspect of the present disclosure,
[0037] FIG. **9** shows a cut view of the clip ring of FIG. **7**,
[0038] FIG. **10** shows a clip ring usable in a mechanical fitting according to another aspect of the present disclosure with a pipe,
[0039] FIG. **11** shows details of the clip ring of FIG. **10**,
[0040] FIG. **12** shows of a tool according to present disclosure,
[0041] FIG. **13** shows a tool according to another aspect of the present disclosure,
[0042] FIG. **14** shows a tool according to another aspect of the present disclosure,
[0043] FIG. **15** shows a method of assembling a pipe with a mechanical fitting, according to an aspect of the present disclosure.
DETAILED DESCRIPTION OF THE INVENTION
[0044] The description below is made in reference to union fittings, but this is not limitative, and the following description applies to elbow, tee or reduction fittings which can be manufactured on the same teachings.
[0045] In the figures, identical parts are identified using the same reference numbers.
[0046] FIG. **1** to FIG. **3** shows a mechanical fitting **1** allowing the connection of a pipe **2** according

to one aspect of the present disclosure, and FIG. **2** shows the mechanical fitting **1** with the pipe **2** after assembly.

[0047] The pipe **2** may be for example a pipe for/in a piping system for fluid installations inside or outside buildings. The pipe may be made of PVC based materials such as c-PVC, PVC, and PVC-U piping systems for hot and cold-water installations.

[0048] The fitting **1** may form one side of a connector having similar or another kind of fitting on its other end or side. The fitting may also be provided at a port of a fluidic device for connecting a pipe thereto.

[0049] The mechanical fitting **1** comprises a main body **10** for receiving the pipe. The main body **10** has an inner body surface **15** and defines a pipe insertion space extending in an axial direction, between an open end **11**, where the pipe **2** is inserted, and an inner radial rib **12** extending radially internally from the inner body surface **15**. The inner radial rib **12** is provided to stop the pipe inserted in the main body **10**.

[0050] When the pipe is a multilayered pipe, a joint gasket can be provided at the inner radial rib to protect the multilayer pipe.

[0051] A nut **40** is provided to be moved into the main body **10** in an assembly position when the pipe is inserted into the main body, to fix and maintain the pipe in place.

[0052] As can be seen, the nut **40** is not provided into the entire main body **10** but only in a nut receiving part **18** of the main body **10** extending axially between the open end **11** and a clip receiving part **19**. The clip receiving part **19** houses a clip ring **30** abutting against an O ring **50** in a joint housing **58** defined by an inner radial projection **16** on the inner body surface **15**. During assembly, the nut **40** pushes the clip ring **30** to press the O ring **50** against the inner radial projection **16**.

[0053] As can be seen on the figure, the inner radial projection **16** is axially spaced apart from the inner radial projection **12**. With other words, not all of the pipe length inserted into the fitting **1** will be below the nut **40** and the pipe free end abutting against the inner radial projection **12** is not provided below the nut **40**. This allows taking into account any variations during installation of the pipe, which either may not be fully inserted, or which may have a cross section which is slightly inclined. Indeed, it may happen that during installation of a pipe system, the user cuts the pipe and that the cut is not perfectly straight. This free space between the inner radial projection **12** and the inner radial projection **16** ensures that all the pipe end is compressed, even if the pipe end is not straight.

[0054] The nut **40** has a first nut portion **42** to be threaded into the nut receiving part **18** of the main body **10**. To do so, the nut receiving part **18** has internal threads **17** extending from the open end **11**. The internal threads **17** are provided for threading the nut **40** into the main body **10** when the pipe is inserted and positioned within the main body **10**. The nut **40** has therefore corresponding nut threads **47**. However, threading is an example only, and the nut could slide into the main body.

[0055] The nut **40** has also a nut head part **44** extending axially between the first nut portion **42** and a nut free end **48**. The head part **44** is provided to abut against the open end **11** of the main body.

[0056] The clip ring **30** is illustrated on FIGS. **6** to **9** in a first aspect, and another example of clip ring **130** is shown on FIGS. **10** and **11** in another aspect.

[0057] The clip rings **30**, **130** are designed to maintain the pipe against pullout force due to internal pressure or external load.

[0058] The clip rings **30**, **130** have an inner circumferential boss or protrusion **32**, **132**, respectively, to be clipped into a corresponding groove **4** of the pipe **2**.

[0059] As can be seen on the figures, the boss **32** defines an insertion surface **33**, **133** extending from a pipe insertion end to an inner radial boss surface **34**, **134**, followed by a pull-out surface **35**, **135** on the pull-out side. The geometry of the clip ring differs in the first aspect shown in FIGS. **7** to **9**, and in the other aspect shown in FIGS. **10** and **11**.

[0060] In the first aspect shown on FIGS. **7** to **9**, the insertion surface **33** forms an insertion angle with respect to the inner boss surface **34** at an angle which is wider than a pull-out angle formed by the pull-out surface **35** with respect to the inner boss surface **34**.

[0061] The dimensions of the groove and of the boss depend on the diameter and the operation pressure of pipe, as well as the thickness of the pipe and the material of pipe. For example, the groove may have a depth be in a range of 2 to 4%, preferably 2.5 to 4.0% of the pipe diameter, and/or in a range of 10% to 30% of the pipe thickness.

[0062] Note that the groove could have other shapes such as circular, square, or rectangular shape.

[0063] Finally, the clip ring **30** has a support surface **36** extending axially from the pull-out surface **34**, so that the support surface **36** defines a shoulder which lies against the pipe and allow reinforcing the pipe at the groove zone.

[0064] The clip ring **130** of FIGS. **10** and **11** differs from the clip ring **30** mainly in that the clip ring **130** has two support surfaces **136***a*, **136***b*. A first support surfaces **136***a* extending axially from the insertion surface **133** and a second support surfaces extending axially from the pull-out surface **134**. With this geometry, the boss **132** is in-between two shoulders **136***a*, **136***b* designed to lie against the outer surface of the pipe **2** and allow reinforcing the pipe at the groove zone.

[0065] The clip ring **30**, **130** is an open ring which allows redistributing the axial pullout load into axial and radial load on the body.

[0066] The inner diameter of the clip ring **30** when open is higher than the outer diameter of pipe, to facilitate the assembling and dissembling with the outer groove **4** of the pipe.

[0067] The clip ring **30** is provided to deform and close when the nut **40** is threaded in the main body **10**. When closed, under load, the inner diameter of the clip ring **30** is reduced and becomes equivalent to the outer diameter of the groove **4**.

[0068] It should be noted that, in addition to maintaining the pipe against pullout force, the clip ring **30** also provides a support to reinforce the groove zone under operating pressure. When the nut is fully closed and reaches to the body, this provides an indication of good assembling.

[0069] Finally, the clip ring has an outer conical shape, with an outer conical surface **37**, **137**, with the diameter decreasing from the pipe insertion end to the fitting side. The clip receiving part **19** has a corresponding conical shape, with a diameter decreasing from the threads **47** to the joint housing **58**, designed to cooperate with the ring conical surface **37**, **137**.

[0070] The conical shape helps closing the fitting on the pipe. When the nut slides or is threaded on the nut receiving part **18**, the clip ring **30** under load closes itself around the pipe.

[0071] To distribute the load homogeneously along the pipe and avoid local concentration of the load, the clip ring **30** may not be a full closed ring and can be an open ring instead. The clip ring **30** can be made as an open ring in one part as shown in FIG. **6** with an open channel, or it can be made of several part. The open ring allows a diameter reduction of the clip ring. Preferably, the clip ring is fully closed in the assembled fitting.

[0072] As mentioned above, the clip ring **30**, **130** is designed to cooperate with a groove **4** on the pipe. The skilled person understands that the groove **4** needs to be made on the pipe **2** at a right position.

[0073] It should be understood that the groove can be made in the pipe, preferably on spot or just before assembly. The groove **4** is circular, even if the pipe has some ovalisation. With other words, the groove is an ovalisation compensation tool. This is particularly advantageous when using PVC pipes which are brittle, and which cannot withstand deformation. Instead of trying to reshape the PVC pipe to be circular again, the fitting of the present invention allows using a circular groove, made in the pipe on the spot.

[0074] Finally, the pipe **2** as seen on the figures is preferably chamfered for facilitating the insertion on the fitting **1**.

[0075] A method of assembling a fitting and a pipe is shown on FIG. **15**.

[0076] First, the pipe is inserted the pipe into the fitting main body and the clip ring, preferably up to the inner radial rib **12**, at step S**1**. The pipe pushes the clip ring against the joint, and when the

clip ring cannot move axially anymore, the pipe groove and the clip ring can engage each other.
[0077] It is possible to clip the clip ring on the pipe before insertion and then insert both the pipe with the clip ring in the fitting. In both cases, the boss of the clip ring is inserted into the groove of the pipe.
[0078] Then, the nut is moved inside the main body, by sliding and/or threading the nut in the main body to close the clip ring **30** around the pipe, at step S**2**, until the completion of the connection (step S**3**).
[0079] The groove is preferably made on site, and it is possible to form simultaneously the outer circular groove on the pipe and chamfering, and possibly correct, the pipe end to be inserted.
[0080] It is advantageous to make the groove and tube chamfering can be done by the same tool.
[0081] FIGS. **12** to **14** shows different examples of tools.
[0082] FIG. **12** shows a tool **80** with a tool body **81** which can be mounted on the pipe end, the tool body extending axially between a pipe insertion end and a pipe stop **82** to stop the pipe end. The pipe stop **82** provides a radial contact surface for the pipe end and is designed to be in contact with the cross section of the pipe end. The pipe stop **82** is therefore perpendicular to the tool body **81**.
[0083] The tool **80** is provided with chamfering blades **83** for chamfering the pipe end. The chamfering blades **83** extend axially from the pipe stop **82**. The tool **80** has also a groove cutter **84**, at an axial distance from the pipe stop **82**. The axial distance between the pipe stop **82** and the groove cutter **84** corresponds to an axial distance between the pipe stop **12** and the boss of the clip ring **30**, **130** of the mechanical fitting.
[0084] As seen on FIG. **12**, corrective blades **85** are provided on the pipe stop **82** for adjusting the cross sectional cut of the pipe end. Indeed, during installation, the user may cut the pipe and once cut, the pipe end may have a cross section which is slightly inclined, not perfectly straight, or may have some marks resulting from the cut. This is especially the case for large diameters when the tube is cut manually. The corrective blades **85** can correct the cross section of the pipe end abutting against the pipe stop, and the end surface of the pipe end. It is also advantageous to make the tube chamfering, and a corrective tool of the pipe end to be inserted in the same tool.
[0085] FIG. **13** and FIG. **14** show two other examples of tools **180**, **280**, respectively. The tools of FIG. **13** and of FIG. **14** can adapt to different range of pipe diameters. In the examples shown, the tools **180**, **280** have a tool body **181**, **281** which can be mounted on the pipe end, the tool body **181**, **281** extending axially between a pipe insertion end and pipe stops. The tools have a first pipe stop **182***a*, **282***a* providing a first stop surface for a first diameter, and a second pipe stop **182***b*, **282***b* providing a second stop surface for a second smaller diameter. The tools **180**, **280** can therefore adapt to two pipe diameters, as an illustrative example only.
[0086] Chamfering blades **183**, **283** and corrective blades **185**, **285** are provided. On FIG. **13**, there is only one chamfering blade and one corrective blade (not visible). On the other hand, the tool of FIG. **14** has four chamfering blades and four corrective blades, for each diameter. Increasing the number of blades allow to avoid rotate a tool 360° to make a complete chamfering. For example, with four chamfering blades, a rotation of 90° with make complete chamfering. The number of chamfering blades and corrective blades, as well as the number of groove cutters can be adapted according to the needs.
[0087] The chamfering blades and/or the corrective blades may be symmetrical to allow rotation in both directions of the tool, depending on user's preferences.
[0088] The tool **180** of FIG. **13** and the tool **280** of FIG. **14** are further provided with more than one groove cutter **184**, **284** on the main body **181**, **281** of the tool. It is possible to make and adapt the dimensions of groove for different pipe diameters, using different groove cutters, such as on FIG. **13** showing two groove cutters.
[0089] The blades of the groove cutters can be radially displaced to adjust to the depth of the groove to be made.
[0090] It should be noted that, on FIG. **13** and the tool **280** of FIG. **14**, the groove for small

diameter pipe can be done by a smaller groove maker installed on the tool, because the groove dimension is different. It is also possible to use a same groove dimension for more than one diameters, for the close diameters.

[0091] The tool **180** of FIG. **13** and the tool **280** of FIG. **14** have handles **187**, **287** to facilitate the rotation of the tool. This is helpful for big pipe diameters.

[0092] In one embodiment, the position of the grove cutter **84** can be adjusted, to adapt to the mechanical fittings and pipe used.

[0093] The clip-ring **30** is designed to push the O-ring **50** to provide sealing. The internal surface features of clip-ring will be stacking in the reshaped groove on the pipe and the conical outer surface of clip-ring will help the pipe to adjusting the pipe holding and redistribute the axial pullout load into axial and radial load on the body.

[0094] Thanks to interface between the joint **50** and the pipe, the joint **50** will deform against the main body in one side and against the rigid pipe on another side, thereby insuring the tightening function.

[0095] The joint **50** can have an internal triangle shape, to facilitate the pipe insertion.

[0096] The joint is a rubber joint preferably made of reinforced polymer.

[0097] The nut **40** is preferably made of polymer.

[0098] This technology has significant impact to replace solvent cement connection for CPVC, PVC, and PVC-U pipe.

[0099] The advantage of the friction mechanical fitting is easy to connect, and the network system become operational quickly.

[0100] The foregoing description of the preferred embodiments of the disclosure has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed, and modifications and variations are possible in light of the above teachings or may be acquired from practice of the invention. The embodiment was chosen and described in order to explain the principles of the invention and its practical application to enable one skilled in the art to utilize the invention in various embodiments as are suited to the particular use contemplated. It is intended that the scope of the invention be defined by the claims appended hereto and their equivalents.

## Claims

**1**. A mechanical fitting for connecting pipes or for attaching a pipe, comprising: a main body extending axially from an open end and defining an insertion space for a pipe, a clip ring designed to close over a pipe, a nut movably mounted at least in part into the main body for tightening the clip ring, wherein the clip ring is arranged inside the main body for being contactable by the nut and has an inner circumferential boss or protrusion, to be clipped into a corresponding outer circumferential groove of the pipe.

**2**. The mechanical fitting of claim 1, wherein the boss has an insertion surface extending from a pipe insertion end to an inner radial boss surface followed by a pull-out surface, wherein the insertion surface forms an insertion angle with respect to the inner radial boss surface wider than a pull-out angle formed by the pull-out surface with respect to the to the inner boss surface.

**3**. The mechanical fitting of claim 2, wherein at least one of the insertion surface and the pull-out surface is provided with inner grooves.

**4**. The mechanical fitting of claim 1, wherein the clip ring has at least one support surface extending axially from at least one of the pull-out surface and the insertion surface, forming at least one shoulders designed to lie against the pipe outer surface of the pipe when the pipe is inserted.

**5**. The mechanical fitting of claim 1, wherein the clip ring is an open ring, wherein an inner diameter of the clip ring when open is larger than an outer diameter of the groove, and with an inner diameter of the clip ring when closed is equivalent to the outer diameter of the groove.

**6**. The mechanical fitting of claim 5, wherein an inner diameter of the clip ring when open corresponds to an outer diameter of a pipe inserted therein.
**7**. The mechanical fitting of claim 1, wherein the clip ring has an outer conical shape, with an outer conical surface, with the diameter decreasing from the pipe insertion end to the fitting side, and wherein the clip ring is housed in a clip receiving part of the main body, the clip receiving part having a corresponding conical shape.
**8**. The mechanical fitting of claim 1, wherein the nut has a first nut portion and a nut head, with the first nut portion to be mounted inside a nut receiving part of the main body the first nut portion having nut threads to be threaded into internal threads of the nut receiving part, and the nut head extending from the first nut portion, the nut head being provided to abut against the open end of the main body.
**9**. The mechanical fitting of claim 1, wherein a joint is provided inside a joint housing portion in the main body located adjacent the clip receiving part.
**10**. The mechanical fitting of claim 1, wherein at least one of these features is fulfilled: a. the nut is made of fiber reinforced polymer, polymer matrix, or composite, b. the main body is made of fiber reinforced polymer, polymer matrix, or composite, c. the ring is made of polymer, fiber reinforced polymer, polymer matrix, or composite.
**11**. Method of assembling a fitting according to claim 1 and a pipe provided with an outer circumferential groove, comprising: one of inserting the pipe into the fitting main body and the clip ring or inserting both the pipe with the clip ring mounted thereon into the fitting main body, moving the nut in the main body, by at least one of sliding and threading the nut in the main body to close the clip ring around the pipe, comprising inserting the boss of the clip ring into the groove of the pipe.
**12**. Method of assembling a fitting according to claim 1 and a pipe, wherein the groove is provided by a step of cutting the pipe to form the groove.
**13**. Method of claim 12, comprising the steps of forming simultaneously the outer circumferential groove on the pipe and at least one of chamfering the insertion end of the pipe and correcting the insertion end of the pipe.
**14**. Method of claim 12, comprising mounting a tool having a main hollow body over the pipe with a pipe stop to stop the pipe end, the tool having a groove cutter, at an axial distance from the pipe stop, the method comprising adjusting an axial distance between the pipe stop and the groove cutter to correspond to an axial distance between the pipe stop and the boss of the clip ring of the mechanical fitting.
**15**. Tool usable in the method of claim 12, wherein the tool has a main hollow body to be mounted over the pipe, with a pipe stop to stop the pipe end, the tool having a groove cutter, at an axial distance from the pipe stop, wherein the axial distance between the pipe stop and the groove cutter can be adjusted to correspond to an axial distance between the pipe stop and the boss of the clip ring of the mechanical fitting.
**16**. Tool according to claim 15, having at least one of a chamfering blade extending from the pipe stop for chamfering the pipe insertion end, and a corrective blade at the pipe stop for correcting a cross section of the pipe end, and a pair of arms extending outwardly from the main body.